A Smart Safety System for Makerspaces: Final Report

Senior Design Project

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Executive Summary

Industrial bandsaws pose a significant risk to both novice and experienced operators due to inadequate awareness of hand proximity to the saw blade from lack of practice or comfort with operation, respectively. Current available safety features primarily rely on physical guards which often limit blade visibility. This project proposed the development of a safety device that serves as an addition onto a bandsaw to alert the user whenever their hand is too close to the blade. The device detects when a user's hand enters a hazardous zone near the blade, provides immediate visual alerts, is easily attachable to various bandsaw models, allows for sensor calibration, and operates reliably in industrial environments. This innovation aimed to increase safety with an easy-to-install, unintrusive, and intuitive system.

Key challenges included balancing cost with performance, maintaining a compact and stable design, and ensuring the system did not interfere with tool operation. The system's main functions are sensing hand proximity, flexible and stable mounting, independent power operation, and immediate signaling of hazardous behavior. After evaluating different technologies, a computer vision AI model was chosen for hand detection due to its high accuracy. The model monitors bounding boxes around detected hands and triggers a bright red LED alert when a hand crosses into the blade's cutting path. Powered by a battery pack, the device offers easy installation without external power reliance. An integrated power management system, combined with an arming indicator, provides users with operational feedback.

Key Performance Specifications that dictate design choices are listed below:

- Device is capable of sensing unsafe operation at least 12" away
- Device has continuous power for 6 hours (one operational day) of heavy use
- Device has total dimensions of at most 12" x 4" x 4"
- Device can produce at least 750 lux (85 lumens) of light for visual warning
- Device has a response time faster than 109 milliseconds
- Bill of Materials should be at most $60

Performance validation focused on measuring power consumption and battery longevity, testing the effectiveness of the calibration system under varied conditions, and testing the effectiveness of the hand recognition system under varied conditions. The device was engineered to optimize power consumption, heat management, and structural stability. Current draw was measured under different operating conditions, and battery capacity needs

were calculated based on high-frequency use. Milwaukee M18 XC batteries were prioritized, meeting or exceeding required capacity specifications after voltage step-down. The structural analysis showed sufficient strength and safety margins for static loads and vibrations, while the device casing, made from PLA, provided durability and passive cooling. Active cooling using a fan and heatsink ensured the Raspberry Pi stayed below critical temperatures during normal operation.

The final design integrates a modular battery receiver with passive ventilation, a Raspberry Pi 5 as the main processor, a two-axis gimbal-mounted camera for hand detection, and a programmable LED alert strip. The device case was updated to improve protection, cable management, and heat dissipation. The software operates using an event-driven architecture with real-time performance optimizations, combining vibration sensing, slit detection, hand tracking, and LED alerts. Although it does not fully meet initial specifications concerning hand proximity alerts from all directions, it effectively enhances user safety around the cutting path.

Manufacturing considerations suggested shifting from 3D printing to injection molding for scalability and cost reduction. Component cost optimization, particularly replacing the Raspberry Pi with a lower-cost processor, was identified as a future improvement. Risk assessment emphasized minimizing false negatives, handling hardware failures, and encouraging user compliance without interfering with machine operations. The system enhances safety awareness through visual alerts without physically controlling the bandsaw, reducing legal liabilities and encouraging broad user acceptance. Overall, the design successfully balances functionality, safety, manufacturability, and cost-effectiveness, positioning itself as an assistive safety mechanism rather than a mechanical safeguard.

This document summarizes the team's process of narrowing the project scope to a bandsaw-focused design targeting laceration risks and progressing through prototyping. While the Gantt Chart timeline was mostly followed, prototyping started more gradually. The team refined both hardware and software, improving precision, processing speed, and incorporating features like self-calibration and dust shielding. Early challenges with communication and workload distribution were resolved, enabling steady progress and ensuring the final product met target specifications by the Design Expo.

1. Introduction and Background

Design Problem, Motivation, and Need:

Industrial bandsaws are widely used in manufacturing, woodworking, and educational environments, yet they pose a significant safety hazard. A persistent issue in engineering and industrial sectors is the accidental cutting of users' hands due to a lack of awareness regarding the distance between their hands and the blade. Poor safety practices are particularly common among novice users, who may lack the experience needed to operate the bandsaw safely. To address this issue, the development of a safety device is proposed to alert users whenever their hands move too close to the blade. This innovation aims to increase safety, particularly for inexperienced users, by enhancing situational awareness and reducing the likelihood of severe injuries.

Intended Use and Operating Environment:

The proposed safety system is designed for compatibility with a wide range of industrial bandsaws, ensuring adaptability across various workspace environments. The system continuously monitors user movements and provides real-time alerts when hands are detected in close proximity to the blade. It is intended primarily for use in educational institutions' makerspaces, where novice users are prevalent and bandsaw safety remains a critical concern. By integrating the system into normal bandsaw operations, the design aims to enhance safety and awareness without disrupting the user's workflow or interfering with standard machine functions.

Desired Product Functions and Features:

The primary function of the device is to monitor the cutting area and detect when a user's hand approaches the bandsaw blade. The system explores various methods to establish and monitor a virtual boundary around the blade, enabling real-time detection and warning capabilities. The device is designed to be compact, lightweight, and easy to mount on the bandsaw without impairing the normal operation of the bandsaw. Furthermore, the mounting mechanism is intended to be adjustable and flexible, allowing the device to accommodate different bandsaw models and configurations commonly found in industrial and educational settings.

Value Statement and Benefits:

At an affordable price, the proposed device significantly reduces safety concerns for bandsaw users in educational makerspace environments. It also helps users develop greater awareness and caution during bandsaw operation, thereby lowering long-term risks. The system's adaptable design ensures compatibility with a variety of machines without requiring extensive modifications to the bandsaw itself. Stakeholders, including students, workshop supervisors, and industry professionals, benefit from a safer, more user-friendly working environment.

Technical Issues, Challenges, and Opportunities:

Developing a cost-effective system poses several challenges. A major hurdle involves managing the balance between manufacturing costs and selling price while maintaining high safety standards. Careful attention must also be given to the weight of mechanical and electronic components to ensure system stability without compromising usability. Sensor placement presents another challenge, as the system must avoid obstructing the user's visibility of the blade. Since bandsaw workspace dimensions vary across brands, the system must offer sufficient flexibility to fit different models without interfering with normal operation. Addressing these challenges requires balancing affordability, buildability, functionality, and user convenience in the final design.

Potential Solutions and Proof of Concept:

To address sensor concerns, the design explores a range of solutions, including the use of computer vision to detect hazardous hand proximity. Various mounting methods, such as magnetic attachments, are evaluated to create a stable and removable system. Software development focuses on resolving key issues such as establishing reference points for self-calibration. Power source alternatives are also assessed by calculating and comparing specifications for available options. These strategies are integrated into the early prototype phase to evaluate their usability and performance in realistic workspace conditions.

Document Overview:

The remainder of this document covers the key aspects of the design and development process for the Safety System. It begins with a review of existing products and patents related to bandsaw safety. Relevant codes and standards that guide compliance and ensure effectiveness are then examined. A stakeholder matrix outlines the primary beneficiaries and users of the system. Following this, a specification sheet and market research analysis provide the foundation for design requirements. The document continues with a detailed discussion of design concept ideation, presenting various approaches considered during development. An overview of the function tree and conceptual development follows, including an evaluation of different safety system concepts. The feasibility of these concepts is assessed using a morphological chart, leading to a preliminary concept selection with justification. Engineering, an overview of the prototyping phase and final design, manufacturing and commercialization, and risk reduction are subsequently covered. The document concludes with key findings and recommendations for future improvements and system implementation.

2. Existing Products, Prior Art, and Applicable Patents

Machine tools represent an integral part of the educational experience for emerging engineers. However, powered equipment such as vertical bandsaws presents significant risks of accidents and injuries. To address these safety concerns, several products have been developed to reduce the likelihood of accidents and protect users. Examining existing solutions that enhance equipment safety helps establish boundaries and inform the ideation process for the new system.

Bandsaw operation carries the inherent risk of accidental collision between the user's hand and the running blade. According to a study from the Occupational Safety and Health Administration (OSHA), approximately 11.5% of workshop accidents involve bandsaws. Several commercial products address this issue. For example, Guardian, a safety technology company, develops a bandsaw equipped with a 3D vision system capable of detecting gloves near the blade [11]. Upon detection, the blade halts without damaging the machine, significantly reducing the risk of user injury during operation.

The concept behind Guardian's 3D vision technology is patented under US8390821B2 [12], as shown in Figure 1. While most motion-detection systems create a two-dimensional simulation of their environments, this invention incorporates three cameras to construct a three-dimensional simulation, thereby improving detection performance and enhancing overall system effectiveness.

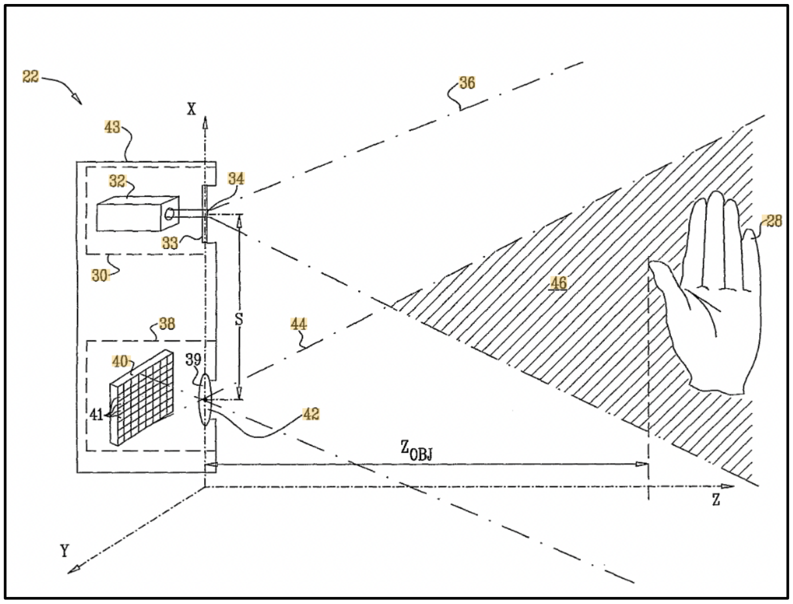

Figure 1: Patent US8390821B2 of 3D vision detection system

Changing the orientation of the machine tool is another way to protect the user from injury. Under patent US9969014B2, a table saw is developed to detect dangerous conditions and retract the blade away from its operational location. This invention consists of a detection circuit and a fusible member that work together to partially retract the blade. Unlike the Guardian bandsaw system, which emphasizes halting the blade, this design prioritizes relocating the hazardous element to a safer position. This concept succeeds in preventing accidents from occuring, however, it is an intrusive system that cannot be easily applied to most commonly found bandsaws. Our solution must be applicable to most vertical bandsaws without the need of installing internal components that alter the machine's operation.

Operating a bandsaw carries a substantial risk of accidents, particularly when used by inexperienced individuals in educational settings. Research into existing solutions for improving machine tool safety provides greater clarity for the ideation process of the proposed system. The reviewed products conditions and timely communications of warnings to the user. Guardian's product does well to protect the user by using a color-based identification system, however, a similar processing system may not be applicable to a makerspace considering that skin color and wood color often overlap. Similarly, the retractable table saw is a sufficient way of protecting users from unforeseen incidents with dangerous blades. Unfortunately, it is also a method that involves deep integration with the machine it is applied to and removes its ability to be applied to another system. Our system must adhere to the client's needs of adjusting a user's lack of supervision while also refraining from becoming an intrusive system that is also capable of being installed to most commonly found bandsaws. Building upon these insights, the proposed solution focuses on developing an intuitive, adaptable device that can be installed on a wide range of bandsaws while providing effective safety alerts without requiring permanent modifications to the equipment.

3. Customer Requirements and Engineering Design Specifications

Low Interest, High Power	High Interest, High Power
Funding Sources Bandsaw Manufacturers	Sponsor of the Project (Professor) Capstone Team Members Capstone Team Collaborators
Low Interest, Low Power	High Interest, Low Power
Component Suppliers Non-Operating Students	Bandsaw Users in Makerspaces Lab Supervisors

Figure 2: Stakeholder Matrix for Bandsaw Makerspace Project

A stakeholder matrix is conducted to analyze the key stakeholders for this project, allowing project members to prioritize communication efforts and design considerations. A critical stakeholder is the project sponsor, who defines and presents the original problem statement and provides ongoing guidance regarding project goals and expectations. Bandsaw users in general makerspaces, as well as lab supervisors in these environments, represent additional important stakeholders. Although these users and supervisors are directly impacted by the installation of the system, they do not exert high influence over the technical design and engineering of the device. This distinction makes it essential for the project team to consult regularly with bandsaw operators, supervisors, and the sponsor to ensure the development of an accurate problem statement, a clear understanding of user needs, and a focused set of functional requirements and specifications.

Additionally, bandsaw manufacturers are identified as important stakeholders. Since the safety device is designed specifically for bandsaws, manufacturers influence the project through the standardized materials, designs, and dimensional specifications of the tools available on the market. These constraints inform key design decisions and ensure compatibility with the intended range of equipment.

The original list of user needs for this project is created through interviews and discussions with key stakeholders. As the design/prototype process progresses, the list is updated and refined to reflect an evolving understanding of the target market and project objectives:

- The device must guide users away from continuing to move their hands into close proximity with the bandsaw blade.
- The alert system must provide clear and immediate warnings.
- The installation and operation must be intuitive and compatible with the majority of educational bandsaws commonly found in makerspaces.
- The device must not interfere with proper operation of the bandsaw.
- The device must include a reliable and compact power source that allows for intuitive maintenance.
- The detection system must automatically calibrate the monitoring window around the bandsaw blade after initial installation without requiring user intervention.

These updated user needs serve as direct guidelines for developing a robust, human-centered device that incorporates the primary safety function while addressing specific design challenges. Based on the identified needs, a specification sheet is developed and continuously updated to provide a quantitative representation of the performance metrics the device must achieve. The full specification sheet, including the date of each specification's creation or revision, the associated validation protocol, and the responsible Capstone team member, is provided in the Appendix as Table A1.

Table 1: Specification Sheet For Bandsaw Makerspace Device [2-5, 14-22]

Specification Sheet For Makerspace Safety System	**Issued: 4/25/25**
Requirements	**Source**
Physical Characteristics	
Device edges have 2mm radius fillet	ANSI/HFES 100-2007 Section 6.2.1.7
Device footprint is at most 12"x4"x4" (LxWxH)	Minimum Dimensions for 1.5+ HP Bandsaws on Market
Device/System is under 51 pounds	NIOSH Lifting Equation with 2 Hands
Operation	
Auditory sensing alerts are at least 110 dB	ISO 7731 Ergonomics
Device is capable of producing at least 750 lux (85 lumens) of light for safety	OSHA Factories and Workshops Lighting Requirement
Safety boundary is a minimum of 3" clearance around blade in all directions	Kansas City Woodworkers Guild Safety Guide
Device has continuous power for at least 6 hours of high-frequency use	Invention Studio hours
Sensing	
Device is capable of continuously sensing unsafe operation up to 12 inches away	Doall bandsaw dimensions (from a GT facility)
Device has a total response time of less than 109 ms	Minimum human reaction time data
Device processes a maximum of 50 FPS	Raspberry Pi Camera 3 Specification
Human Factors	
Device has no exposed electrical components.	IEC 60204-1 (Safety of machinery - Electrical equipment of machines)
Device needs to be maintained/recharged at most once every 6 hours	General makerspace guidelines, Invention Studio hours
Miscellaneous	
BOM cost can be at most $60	Market Research (30% of 20% of Cheapest 1.5 HP Band Saw)

The following sections explain each specification and their respective sources in more detail.

Physical Characteristics:

The physical specifications for the device establish basic restrictions on the design of its main body. ANSI/HFES 100-2007, a standard titled *Human Factors Engineering of Computer Workstations* [2] published by the Human Factors and Ergonomics Society, provides ergonomic guidelines applicable to this project. Section 6.2.1.7 of the standard states that any hard surface edges that come into contact with the user during normal operation should have a minimum radius of 2mm [2]. Although the device is not designed for a computer workstation, applying this specification promotes safer handling during installation and maintenance, reducing the risk of user injury from sharp edges.

Based on the targeted market outlined in the Market Research section, the minimum dimensions for a 1.5+ HP bandsaw allow for a maximum device footprint of 12"x4"x4" (LxWxH). The length dimension is determined from the throat depth of a reference bandsaw [3] representing the smallest sizes in the specified market. The width dimension is based on visual inspection of the throat width relative to the known cutting surface dimensions, and the height is drawn from the available headroom in the bandsaw throat when the blade is adjusted to maximum cutting depth. These dimensional constraints ensure that the device does not interfere with normal operation, thereby upholding the fourth identified user need regarding unobstructed functionality.

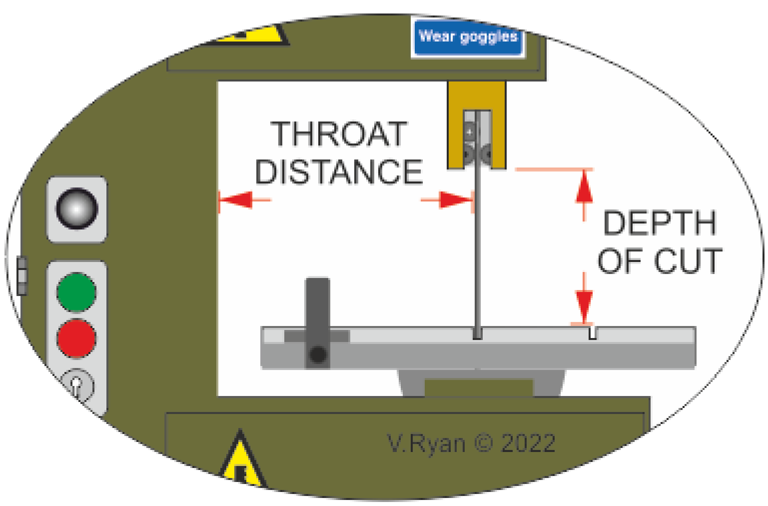

Figure 3: Bandsaw dimension terminology [13]

The National Institute for Occupational Safety and Health (NIOSH) is a federal agency responsible for making recommendations to prevent work-related injuries. Based on the NIOSH Lifting Equation

[5], under ideal conditions, an individual can safely lift a maximum of 51 pounds without increased risk of injury. Because the device is designed for one-time installation with a built-in calibration system, this value establishes an upper weight limit for the device design. It is important to note that this specification remains subject to revision based on future engineering experimentation and testing, including evaluation of the magnetic mounting weight limits and potential injury risks as outlined in the Engineering Analyses section. These studies may lead to a substantial reduction in the final allowable weight.

Operation:

The operation section defines specifications for how the device should perform based on the identified user needs. To ensure that alerts are clear and immediate, any auditory warning alerts included must reach at least 110dB, which is 15dB greater than the ambient noise generated by a bandsaw within the target market, according to the ISO 7731 Ergonomics standard [14]. Furthermore, the OSHA Factories and Workshops Lighting Requirement [15] mandates that factories and workshops maintain a minimum illuminance of 750 lux, equivalent to 85 lumens on a bandsaw cutting surface. This establishes a minimum threshold for the brightness of any visual warning signals incorporated into the device.

Market research identifies that the most critical safety hazard involves users moving their hands too close to the blade. Based on this observation, a specification defines a minimum safety clearance of 3 inches around the blade, as recommended by the Kansas City Woodworkers Guild [16] safety guide—a non-profit educational woodworking shop based in Kansas City. It is important to note that this specification remains subject to revision as ongoing conversations and interviews with key stakeholders, including bandsaw users and makerspace supervisors, continue to refine user needs and operational requirements.

Sensing:

The initial specifications are based on the dimensions of a Doall bandsaw [17] located at the Georgia Tech facilities. Given that the bandsaw bed measures 24 inches in length, the camera must be capable of detecting unsafe operations at a maximum distance greater than 12 inches. This distance represents the furthest possible mounting point from the blade along the cutting direction.

To ensure immediate alerts, the device specifies a response time–defined as the interval between detecting unsafe operation and initiating the alert–of no more than 109 milliseconds, based on minimum human reaction time data [18]. Additionally, the camera's maximum frame rate is set at 50 frames per second (fps) [19], balancing detection accuracy with power consumption considerations. This frame rate specification remains subject to adjustment as design development and experimental testing determine whether lower frame rates can maintain adequate performance while improving energy efficiency.

Human Factors:

Several specifications are developed to ensure that human factors are incorporated into the device design. The first specification, sourced from IEC 60204-1, *Safety of machinery - Electrical Equipment* [20], mandates that the device must have no exposed electrical components. This requirement necessitates an enclosed casing and wire management to ensure user safety during installation and operation.

As stated, the Invention Studio operates six hours a day. Based on discussions with supervisors and observations by capstone team members in the makerspace, it's clear that supervisors follow a routine checklist at the start and end of each day. With this insight, a specification is established to emphasize intuitive battery maintenance. The device must support a battery lifespan of at least six continuous hours before requiring replacement or recharging, posing a critical design challenge for the project.

Miscellaneous:

Based on the market analysis detailed in the Market Research section, the Bill of Materials (BOM) for the device should not exceed $60, targeting affordability for manufacturing and potential large-scale production.

House of Quality:

The following figure is a portion of the updated House of Quality, specifically the listed engineering requirements and each relative weight. The full HOQ can be found in the Appendix as Figure A1.

Dimension Constraint	10%
Adequate sensing range	13%
Consistent intersection sensing (safety)	14%
One day of continuous operation	11%
Secure attachment	9%
Low BOM cost	7%
Clear auditory alert	10%
Clear visual alert	10%
Robust, aesthetic infrastructure	8%
Lightweight	8%
Quick sensing and immediate alert	12%
Safe interface with user and bandsaw	8%

Figure 4: Section of House of Quality: Engineering Requirements and Relative Weights

A key outcome observed from this figure is that the engineering specifications and requirements have remained consistent in scope since the beginning of the design phase. The engineering specifications with the highest relative weight include a Adequate sensing range, Consistent intersection sensing (Safety), One day of continuous operation, and Quick sensing and immediate alert. These specifications collectively highlight the main design challenges that have been integral to the project, including a robust method for detection and power delivery solution.

4. Market Research/Potential Impact

By analyzing the bandsaw market and its applications in educational institutions, we can identify key safety concerns and design parameters for an effective safety system. The target users, particularly students in makerspaces, present unique challenges, as both novice and experienced operators are prone to accidents. Given the frequency of machinery-related injuries in industrial settings, a well-designed safety system is crucial.

The bandsaw machine market is currently US $3.2 billion as of 2025. According to Persistence Market Research [8], It is expected to increase to US $4.7 billion in 2032. The growth of the market can be attributed to the bandsaw's accuracy, speed, and ability to produce intricate patterns. The number of bandsaws sold annually in the US can be estimated to be 75,000 units. When comparing the relative size of the number of institutions and schools to the size of the US manufacturing industry, the number of bandsaws purchased annually by educational institutions is approximately 3,000 units.

Initial impressions from market research indicate that the primary target user for a makerspace tool safety system is the student, both novice operators and experienced ones. While a novice operator might be more likely to use a machine incorrectly, an experienced operator might be less cautious and is more likely to disable or disregard safety systems if they are obtrusive to the user experience. In 2020, the Bureau of Labor Statistics reported that, from the over 1 million workers in the American machinery industry, there were 8,280 injury cases involving workers who had taken days away from work (an analogue for students with relatively infrequent use of makerspace tools) [1]. 1,800 of those were hand injuries.

A primary reason that we are focusing on institutions is because, as students, we have access to educational makerspaces and can better understand the problems lying underneath. The operating hours of these institutions help outline the runtime in the battery consumption analysis. The safety system will be designed with in mind the dimensions of specific brands of bandsaws found in the Georgia Tech Invention Studio as well as common brands used by manufacturers. To better understand the design parameters of the safety system, only bandsaw machines that operate at minimum of 1.5 horsepower will be examined during the research stage. This limit sets the standard for vertical bandsaws to ensure they meet certain size proportions. Smaller machines such as bench top table saws operate below this limit of 1.5 HP and will be excluded in the ideation phase. We will also examine machines from multiple companies to ensure the safety system can be applicable to most bandsaws, including Powermatic, DoAll, Dake, etc. The cheapest 1.5 HP bandsaws are roughly US $1000 [21]. If we model the safety system's seller cost to be 20% of the bandsaw's seller cost, and the cost of production to be 30% of the safety system's seller cost, then the bill of materials should not exceed $60.

Understanding the bandsaw market and its use in educational institutions highlights the importance of developing a reliable safety system. Since students in makerspaces vary in experience, both novice and skilled users face risks when operating these machines. Given the prevalence of machinery-related injuries, implementing an effective safety solution is essential. This research will focus on bandsaws with a minimum power of 1.5 horsepower and consider multiple brands to ensure compatibility. By addressing these factors, the goal is to improve awareness in educational settings without compromising efficiency or accessibility.

5. Design Concept Ideation

Function Tree and Key System Overview:

The primary objective is to prevent accidents on a band saw by warning the user when a hand nears the blade's cutting path: a high-risk area. The main function of the device is to guide users away from incorrect operation of the bandsaw. Our function tree, shown below as Figure 5, breaks this down into primary subfunctions to focus on. The most important functions include the following.

- Connecting to the Bandsaw: Ensuring seamless integration with the machine for reliable operation.
- Connecting to a Power Source: Providing continuous and stable power, whether through a direct connection or a portable battery.
- Detecting Unsafe Operation: Identifying when a user's hand is in proximity to the cutting path using advanced sensors.
- Alerting the User: Delivering immediate warnings through visual and auditory signals to prevent accidents.

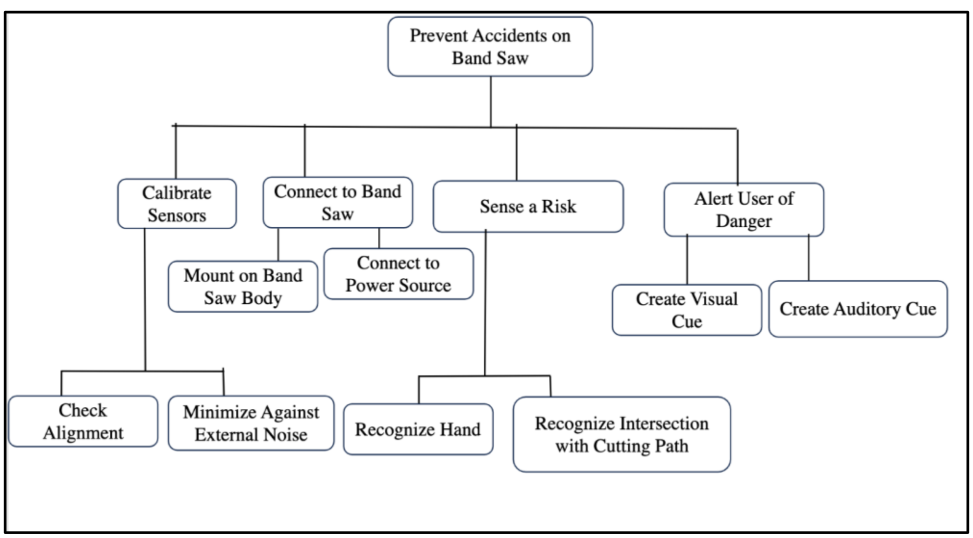

Figure 5: Function Tree, featuring functions desired in Safety System

Conceptual development:

Concept ideas that seemed most plausible for sensing proximity to the blade were an infrared (IR) sensor that triggers if a hand intersects the projected cutting zone box and a camera-based computer vision system. When combined with certain AI models, the latter system could provide a more flexible and possibly more accurate detection system by self-calibrating to the position of the blade and sensing the image of a hand, which would mitigate any false alerts from an IR sensor that could be caused by heat from the blade and workpiece. For warning the user, an immediate visual cue of flashing LEDs and auditory cue of a siren or warning phrase will exist. Power can come from replaceable batteries, which simplifies and increases the transportability aspect of the safety system; however, the caveat with this is the constant battery monitoring. Another option for power is to draw straight from the band saw's motor supply or a wall adapter for constant monitoring. Mounting solutions vary between a magnetic base that can attach anywhere on the machine base, a clamp-style bracket for a more rigid and long-term attachment, and a rubber lined strap for quick placement. Sensor calibration can be acted upon through manual adjustment, semi-automatically if the system includes algorithms to detect the blade and define the "danger zone" automatically.

Design Evaluation via Morphological Chart:

Certain functions need to be addressed in the creation of the Smart Safety System such as sensing the proximity to the blade, warning the user of immediate danger, calibrating sensors, and connecting to the machine body. These functions can be addressed through multiple solutions or design choices through the morphological chart (See Appendix, Figure A2).

More specifically, a morphological chart provides a structured way to analyze different plausible safety system choices in design for each function. Each row of the chart represents a specific function, such as checking alignment/calibration, mounting on the machine body, connecting to power source, alerting the user of system conditions, recognizing hand and intersection with cutting path, and creating audio and visual code. For each function multiple possible solutions are explored. For example, alignment can be achieved using a reference point on the cutting surface or a hardware-based orientation sensing. Noise minimization can be handled through software-based signal processing or hardware insulation. Mounting options include removable hardware like clamps, magnetic attachment for easy repositioning, or adhesive and Velcro for a more permanent fix. Powering the system can involve an external power source or an internal battery for portability. User alerts may be visual, such as LED indicators for power status and proximity warnings, or auditory, such as buzzers for immediate warnings. Hand recognition techniques include infrared sensors, ultrasonic sensors, or computer vision using AI. Intersection detection with the cutting path can also be performed using these same sensing methods. Warning cues can be delivered through flashing lights, displays, electronic sounds, or mechanical alerts. Combinations of each function’s plausible solution with the others gives various design concepts for the safety systems.

Safety System Concepts:

Based on the morphological chart, five distinct bandsaw safety system concepts were developed in hopes of balancing functionality, cost, and usability.

The Computer Vision-Based System uses AI image recognition to track hand placement and cutting path intersection points, ensuring high precision. It minimizes noise through software processing and mounts fixed hardware, powered by an integrated internal battery. A downside of this system could be the significant processing power.

The Lidar-Based Proximity Sensor System relies on lidar sensors to detect hands and reference points. Hardware shielding reduces interference, and a magnetic mount allows for easy repositioning. It runs on an external power source, provides LED indicators, and an audio and light-based warning. This is a cost-effective solution relative to other concept ideations.

The Ultrasonic-Based Safety System employs ultrasonic sensors to detect hands and workpieces, with software-based filtering for more accuracy. It uses Velcro and adhesive mounting and a rechargeable battery for the sake of portability. Alerts include a small display, electronic sound warnings, and a mechanical alarm. This system is user-friendly and flexible.

The Hybrid Vision and Sensor System integrates computer vision and Infrared sensors for redundancy and precision. Hardware insulation minimizes interference, and fixed mounting ensures stability. Powered Externally, it features LED indicators for visual warning. This system offers higher reliability but is pricier.

The Basic Infrared and Mechanical Stop System is a simple, but effective design that employs infrared sensors to detect hands near the blade and a mechanical stop keeps the blade from moving upon detection. Magnetic mounting ensures easy installation, and a built-in battery powers

LED indicators and an audible alarm. This is a relatively cheap option while providing basic safety requirements.

Each concept prioritizes accuracy, ease of use, and cost effectiveness, using different combinations of solutions for each function.

6. Design Concept Selection and Justification

Evaluation of existing concepts and alternatives to finalize a design concept centered around practicality, both in end use and design timeline. A bandsaw-focused design for a makerspace setting required consideration of common elements of bandsaws of the size and quality that would be used in makerspaces like Georgia Tech's Idea Lab. The bandsaw in this space, as well as those in its peer facilities in the same building and similarly priced bandsaws on the market, all featured a design with a C-shaped cutout for the saw blade and cutting surface. Attachment of the detection device to the metal overhang via a mount affixed to a magnet thus appears feasible and excels in the named User Needs of "portability" and "ease of installation". An adjustable housing for the camera will be attached to the mount so it can be angled to best capture images of the cutting surface. The main foreseeable complication comes from vibration-induced motion, which will be mitigated with high magnet strength and a layer of rubber or rubber-like material between the magnet and bandsaw to both absorb vibration and increase friction at the contact surface.

The device will rely on computer vision as its detection mechanism, as it has the most adaptable sensing framework, can detect to close tolerances, and is the least likely sensing tool to present design obstacles when defining the system's method of hazard recognition (given the team's knowledge of prior research projects in this area using computer vision). Furthermore, the timeframe of training the AI model is known not to be prohibitively long, while the timeline of designing and programming a sensor array that fulfills the mission of this project is not easily predictable. The model will be trained to recognize the hand of any user and draw a bounding box around the hand, which will be continuously checked for proximity to the blade and intersection with the cutting path. Both of the latter parameters will be defined by training the model to recognize the position and orientation of the saw blade, which will act as a calibration reference point as well.

When the aforementioned hazardous behavior is detected, the computer will be programmed to turn on a red LED, which will begin flashing if the hand is both intersecting the cutting path and in a close proximity to the blade (as defined by the Specification Sheet).

A battery pack will power the device to continue the theme of prioritizing human factors in the design of the device, which will ensure the greatest chance of an end-product that will be put in to and stay in use. While the drawback of a battery pack is an increase in maintenance requirements (charging or replacing batteries), the alternative implementation of a plug-in power cord creates and extra step in installation and restricts the use of the device to a bandsaw with a nearby available 120 V outlet, in addition to creating a possible tripping hazard if the cord is not properly managed.

As the detection system will be primarily passive, the power demands of the device are expected to be low enough that the battery will need to be serviced infrequently enough to meet the specifications. This assumption, however, relies on a system that limits or triggers elements of the device based on whether it is needed. A vibration-based monitoring system will act to turn the camera and processing capabilities on or off based on whether the bandsaw is in use. This will be powered by the same source as the computer module, but the power consumption of the monitoring system and the idle Pi will be much lower than the fully armed system. Furthermore, another functional consideration is informing the user whether the device is armed, which could be a combined system with active battery-life monitoring (as opposed to an extra system for a cord-based device).

As all considered design concepts were similarly likely to meet the highly-weighted specifications of sensing range and response time in the evaluation matrix (Table 2 below), and the most contentious defining feature of each concept was sensing method, the distinguishing criteria were based on sensing ability and power usage. While a computer vision detection system will draw more power than a lidar, ultrasound, or infrared-based system, the sensing precision of computer vision is on par with any other sensor. The primary distinguishing feature of computer vision, though, is the recognition of a user's hand and determining its bounds, which it performs very well on according to prior research. The other sensing mechanisms, however, may not be able to predictably determine the bounds of the hand at all or may receive interference or false alerts from unpredictable noises in the surrounding environment or heat generated by the blade and the workpiece.

Table 2: An excerpt from Figure A1, the House of Quality, comparing the relative importance of specifications in selecting a design concept

Target	12" x 4" x 4" LxWxH	12" minimum sensing range	3" bounding box threshold	6 hrs continous power at max power draw	Less than 1mm movement p day	<= $60	>=110dB of audio	>=85 lumens of light	>2 mm fillet radius	<51 pounds	<109 ms response time	No exposed electrical components
Max Relationship	9	9	9	9	9	9	9	9	9	9	9	9
Technical Importance Rating	348	476	498	398	310	248	354	354	288	292	416	288
Relative Weight	10%	13%	14%	11%	9%	7%	10%	10%	8%	8%	12%	8%
Weight Chart												

Design challenges with this concept include power demand and software effectiveness. Requisite battery capacity of the device can only be conservatively estimated as an upper bound, as maximum specifications for power draw of components are given by manufacturers, but most of the detection-related activity will happen infrequently for this device. Using a battery with capacity estimated based on maximum figures may generate extra cost, size, and weight, so both implementation of the aforementioned vibration-based power conservation strategy and measurement of actual power demand is underway to minimize these qualities and is discussed below in the Prototyping section. The software challenges involved in designing this system are also an evolving problem, which, to date, includes working around image distortion from camera angle, maximizing processing efficiency to lower power consumption, and refining the computer vision model to measure the boundaries of the hand and reference points precisely. Going forward, these will play perhaps a more critical role in meeting the design specifications than any hardware design challenges.

7. Industrial Design

An important aspect of this project is to create a device that is human-centered, which is highlighted in several user needs covering device installation, operation, and power maintenance. Several considerations informed the design of this device which are highlighted below.

Physically speaking, the chosen concept for this device utilizes magnets to provide an intuitive installation method, eliminating the need to open up any components and making it easy to reposition. The design is split into two parts: the Raspberry Pi 5 Module and the battery module, allowing for adjustable positioning and distributing the weight for easier installation. The specification sheet in Table A1 also requires 2mm radius fillets on all device edges to prevent potential harm during handling. As described in the Mockup and Prototyping section efforts will focus on developing a case design that conceals hardware components, enhancing the device's aesthetic appeal.

This case design features an adjustable camera mount, simplifying the process of directing the camera towards any required orientation for the user's convenience. Furthermore, after consulting with bandsaw users and the project's sponsor, a key objective for this project is to enable the device to automatically calibrate to the orientation of the bandsaw blade upon installation, thereby minimizing the need for ongoing user maintenance.

The power solution for this device, influenced by observations in Georgia Tech's makerspaces, involves using a Milwaukee tool battery—a type commonly found in educational makerspaces. This choice ensures that makerspace supervisors can easily swap out batteries as needed from the existing inventory. Additionally, supervisors can incorporate this task into their routine checklist of activities to perform at the start or end of the makerspace's operational hours.

8. Mockup and Prototyping

An initial or (Phase 0) prototype was created in SolidWorks as shown in Figure 6. This prototype, as described in the Design Concept Selection and Justification section, represents our initial design for the makerspace safety device and was modeled in real life at a 1 to 1 scale.

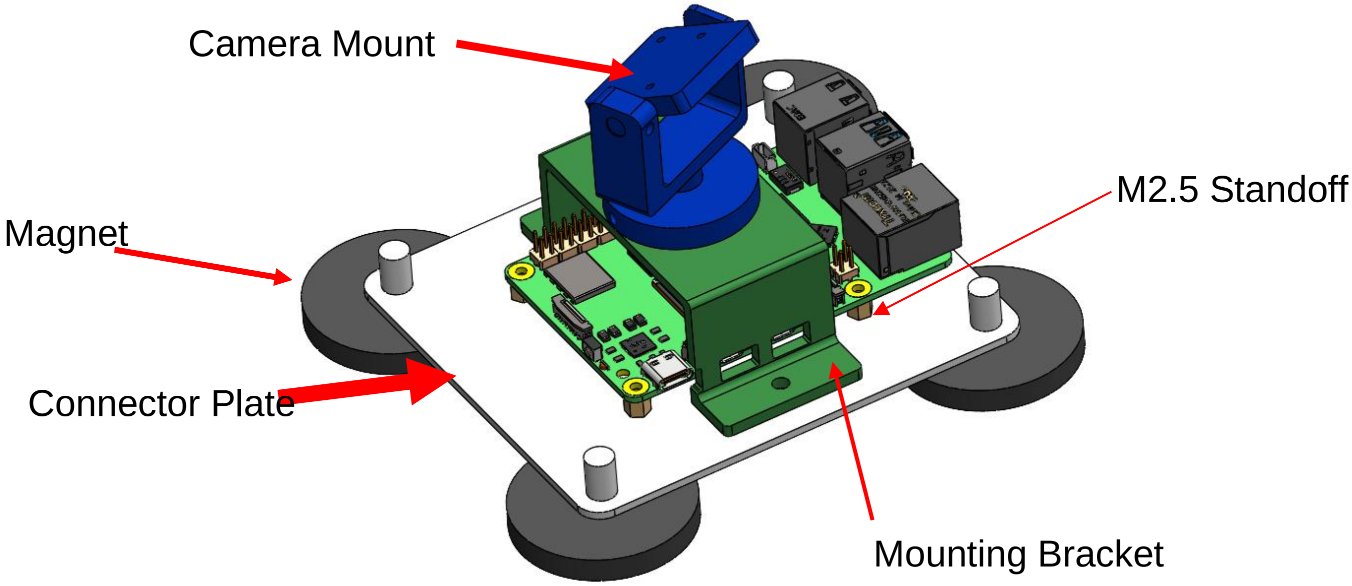

Figure 6: Initial Prototype for Raspberry Pi 5 and Camera (Phase 0)

The primary purpose of this prototype was to test the magnetic mounting solution for the Raspberry Pi 5, determine the optimal orientation of the camera, and establish preliminary mounting dimensions for future design iterations. Please note that the power solution is not shown in this figure, as this concept prototype only includes the mounting solution for the Raspberry Pi 5 and the camera module. A side view of the prototype with dimensions is included in Figure 7, as shown below.

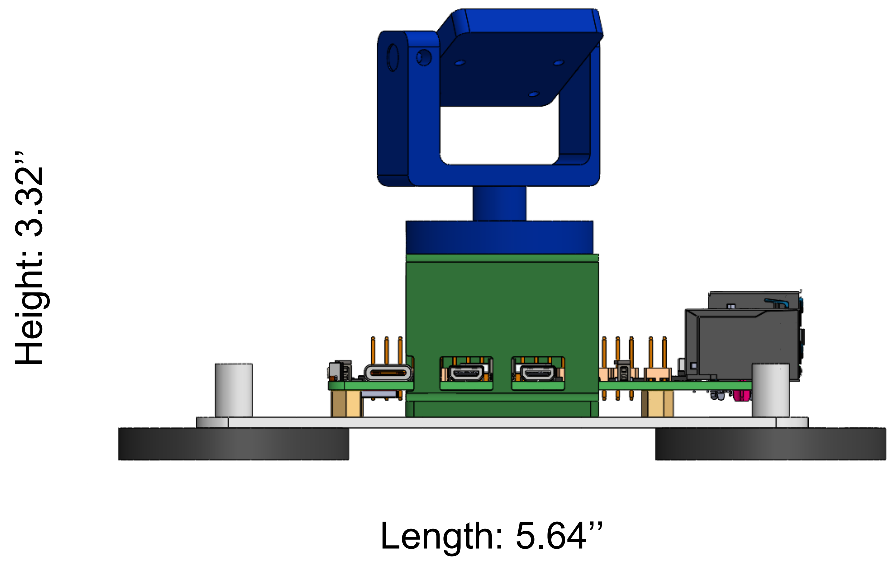

Figure 7: Initial Prototype for Raspberry Pi 5 and Camera, Side View

During the design and construction of this prototype, valuable insights were gained regarding the most effective elements, areas needing improvement, and updates needed. The following list summarizes key conclusions from the construction phase.

- The preliminary design left the Raspberry Pi exposed to dust and particles. This iteration served as a foundation for developing a closed casing solution.
- The number of magnets shown exceeded the necessary holding power for the current prototype, as one magnet is rated at 30 pounds of holding weight. Future design used less magnets to minimize dimensions and cost, though still incorporated multiple for geometrical stability.
- The initial design did not have a cable management system for the camera’s ribbon cable or power cable, violating human factors specifications.
- The M2.5 standoff screws used to mount the Raspberry Pi 5 are secure and easy to use, and can be used to stack additional components in a compact design. These were kept for future design iterations to attach the user alert system or vibration detection system.

Subsequent stages of prototyping incorporated a functional software aspect—meaning it evolved to include being powered with a mounted battery and operational autodetection and calibration functions. Throughout the project, several critical software innovations were landmarks for enhanced system robustness and efficiency. These include:

- Integration of the MPU6050 sensor to enable automatic activation and deactivation of the camera based on tool operation, significantly improving energy management.
- Reduction of the camera frame rate to 30 frames per second to lower CPU usage while maintaining adequate detection fidelity.
- Implementation of selective frame recording, where only frames involving detected hands are saved, minimizing storage requirements.
- Early finalization of the blade slit location, with slit detection performed only during an initial window after activation, thereby reducing continuous processing overhead during operation.

These improvements resulted in a responsive, energy-efficient, and reliable system, while operating within the practical limitations of a Raspberry Pi. The full architecture of the system is described below in the Final Design section.

9. Engineering Analyses

Power Supply:

Manufacturer specifications for a Raspberry Pi 5 indicate that it consumes (at most) 5A and operates at 5V [10] resulting in a 25W max power draw. However, measurements taken with a MakerHawk USB Multimeter indicated that total current draw of the system was, on average, 4.2A when the detection and alert system was active and 1.4A when it was dormant (i.e. only the vibration detection system was active). Ideally, the power would last as long as possible, or at least a full 24 hours, but to generate a preliminary figure, Georgia Tech's Invention Studio hours (6 hrs per day) were used to represent one day of use. To generate a minimum battery capacity specification, power consumption was calculated for high-frequency use over the 6 hour period. Assuming the bandsaw is used once every 3.5 minutes for 30 seconds (as in a 30 second period of use of the device is initiated once every 3.5 minutes), and given that the device draws 4.2A when it is in use and 1.4A when it is not, the device's power source requires 10.8Ah of capacity to meet this specification. Figure 8 below shows a comparison of power consumption with the vibration detection system vs. without, which results in over a 57% decrease in necessary battery capacity.

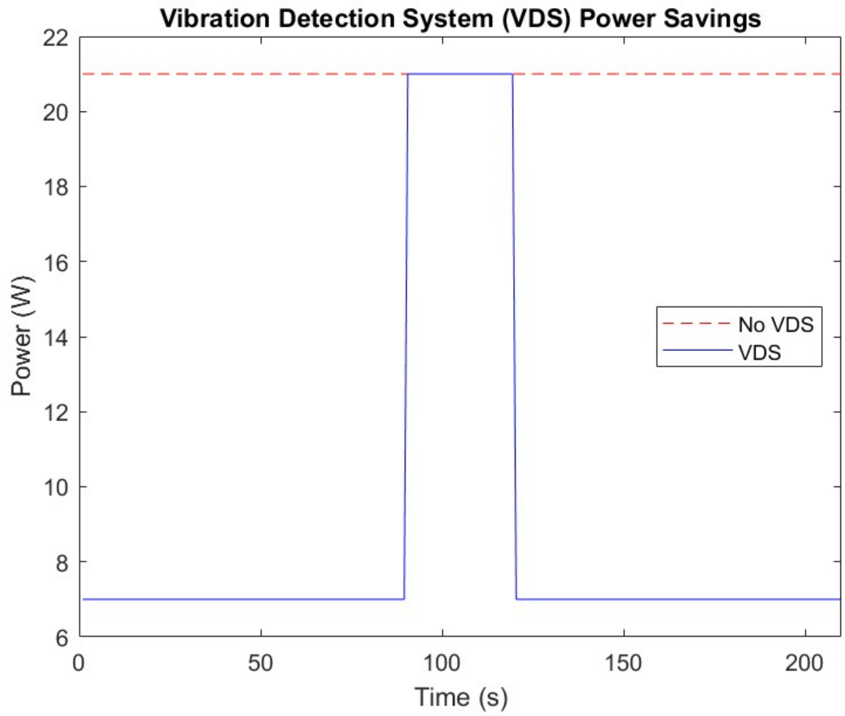

Figure 8: One average cycle of bandsaw use and disuse on a busy day in a makerspace with and without the VDS.

As batteries of the Milwaukee M18 XC are available in Georgia Tech Makerspaces, these were given priority when generating potential battery solutions. These batteries are an 18V DC source, and range from 3.0Ah to 12.0Ah of capacity [7] at that voltage. Using the power equation, the capacity of the batteries when stepped down to 5V can be calculated.

$$P = VI$$

Equation 1: The power equation, using voltage and current.

As the step down to 5V results in an increase of the listed capacity by a factor of 3.6, the M18 XC models extend to ranges of 10.8Ah to 43.2Ah. This, of course, means that any M18 XC battery would meet the minimum power specification and that the largest battery would allow for 24 hours of uninterrupted high-frequency use.

Mounting:

The structural components able to be evaluated on this system are simple. As there are no moving parts, only the statics and failure potential of the hardware need be evaluated, and the only loads on the system are its own weight and vibration. When the system is in use, the weight of the whole system, excluding the battery and its associated hardware, is supported by the device module, which is, in turn, supported by magnets. Thus, certain failure points can be isolated: the magnets (both for the battery module and device module), the connecting hardware, and the connection points within the device module assembly.

The device module's total weight has been measured as less than 1.5 lbs and the largest battery of the selected alternatives weighs 3.4 lbs [7], while each magnet has a holding strength of 30 lbs [24]. Even if the battery module were mounted sideways and the largest possible battery was used (shown below in Figure 9), the 30 lb pulling force would be sufficient, as the center of mass of the battery lies 2.90 inches away from the contact surface, generating a 9.86 in-lb moment about the magnet. It would take 25.35 in-lb to separate the magnet from the contact surface, indicating a margin of safety of 0.61. The battery module would not slide either, as there is a normal force of 41.67 lbf on the contact surface due to the combination of the moment and the magnetic force, and with a coefficient of friction of 0.64 [25], it would require the weight of the battery module to be 26.60 lbs to overcome static friction.

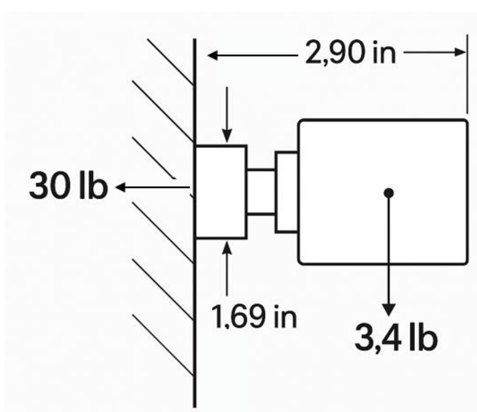

Figure 9: Free body diagram of a sideways-mounted battery module

The material of which the module is printed, PLA, has a tensile strength of about 40 MPa (or 5800 psi) [6]. As the magnets used are connected via M6 steel screws that are welded on one end and screwed into the device module on the other, it can be assumed that the connection would fail in the connection to the module. Even if the stress at this connection is conservatively calculated using one thread as the cross section, the load required to strip one connection would be 76 lbs. Some components of the camera mount are adhered to each other with cyanoacrylate, but the total camera and camera mount weight is less than 25 g. The smallest adhered connection has a cross section of 79 mm^2, and the tensile strength of PLA-PLA bonded cyanoacrylate is 850 psi (5.8 MPa) [9], meaning the required load to break the connection is 102 pounds.

Apart from normal use, the only circumstance where the device would experience external load is during a collision against a solid object after being dropped. As the variables in this circumstance are complex (contact point, deflection upon impact, fracture location, trajectory, collision surface, etc.), drop-testing would be the simplest and most reliable method to evaluate failure potential. Although the resources weren't available to conduct destructive testing during the course of this project, a drop would only occur due to user-handling error or defective assembly of the mounting system, so it was deemed acceptable to exclude this process as it falls out of the designed operation and use of the device.

Heat:

Heat analysis was conducted to ensure that the Raspberry Pi 5 maintained a temperature under 85°C using an official Raspberry Pi active cooler and heatsink module, which is a critical temperature found to cause throttling in Raspberry Pi 5 CPUs [26]. The device uses 5V and draws a maximum of 4.2 Amps when operating, giving a heat generation of 21W from the CPU. In order to find if the heatsink and cooler combination is optimal, real world performance testing history was researched, and it was found that the active cooler and heatsink keeps CPU temperatures at around 56°C at around 10W of heat generation [27]. Assuming an ambient temperature of 25°C, the thermal resistance between the CPU and the surrounding air can be found: $R\theta = (56°C - 25°C) / 10W = 3.1°C/W$. Using this thermal resistance, the temperature of the CPU can be found assuming a constant heat generation of 21W: $\Delta T = 21W \times 3.1°C/W = 65.1°C$ The temperature of the CPU is given as: $T\_CPU = 25°C + 65.1°C = 90.1°C$.

While this temperature exceeds the critical temperature given earlier, this analysis assumed constant operation drawing 4.2 Amps, which is impractical for the Raspberry Pi 5. The use of a VDS allows the Raspberry Pi 5 to stay within the critical temperature.

The battery receiver case, as shown in Figure 10, features ventilation slits for a passive cooling system rather than an active cooling system. An active cooling system is recommended when internal temperatures are between 80º-90ºC. The wiring in the case barely reaches 55ºC, and therefore, it is sufficient for a passive cooling system to be implemented.

10. Final Design

Hardware:

Figure 11 shows the exploded view of the final design, excluding the power module. The design for the power system includes the use of a Milwaukee M18 XC tool battery held in a magnetic casing allowing for easy installation/replacement, as shown in Figure 10. The receiver snaps onto the Milwaukee battery and the combined part slides into the battery receiver case. The receiver case contains the step down converter which safely transforms the current provided by the battery into usable voltage for the raspberry pi. The receiver case includes alignment tabs on its top and side surfaces to restrict the battery/receiver into one position. The receiver has the same magnet that is present on the receiver which allows for equal leveling and a smooth installation. There is a sliding floor that can be inserted into the receiver case to ensure the wiring of the power system is tucked away and prevented from interfering with other systems. The receiver also features ventilation slits on both sides of the case. This was done to implement some form of a passive cooling system for the case. An active cooling system, such as an internal cooling fan, would only be necessary if temperatures of the wiring/components exceeded 80º-90ºC, whereas the wiring barely crossed 55ºC.

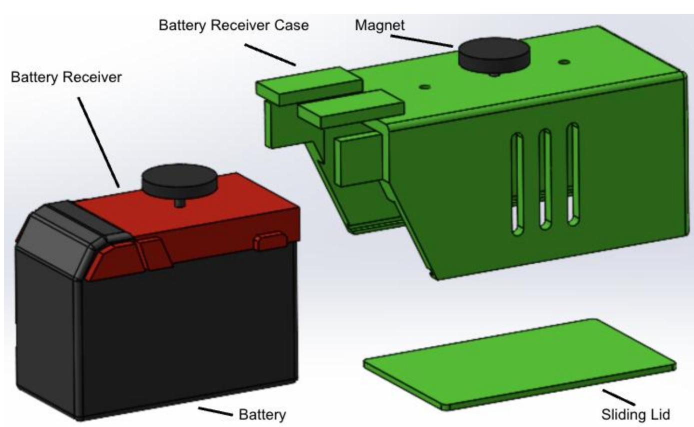

Figure 10: Exploded-view of the battery receiver and receiver case

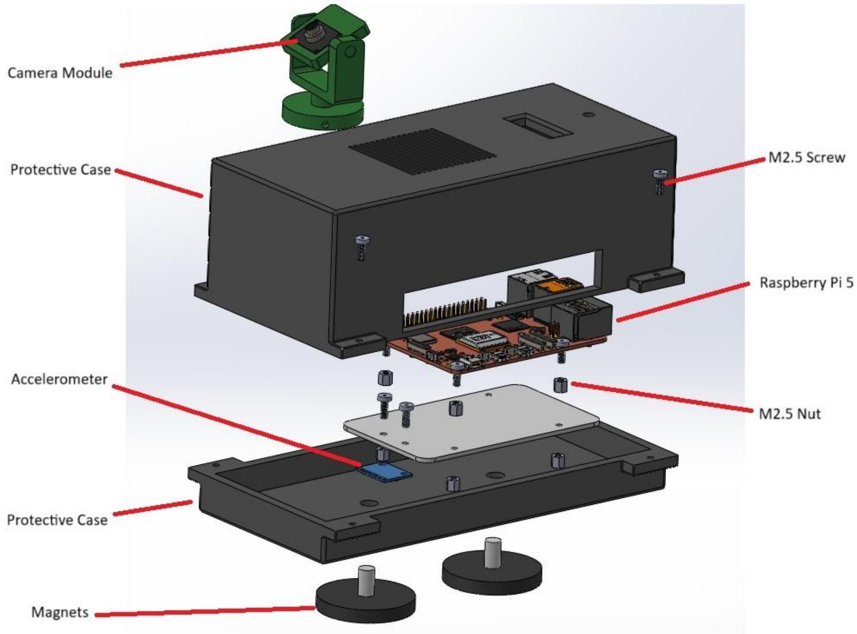

Figure 11: Exploded-view of the device module

The following paragraphs give a detailed summary outlining components of the final design.

Raspberry Pi 5: The Raspberry Pi 5, which handled much of the processing in our initial prototype, remains the primary module performing the heavy processing work in this updated design. This module is central to the camera operation, vibration detection system, and LED system. The Raspberry Pi 5 uses code that is uploaded through Python software that operates all of the systems. The module also includes an active cooling fan and heatsink device that integrates seamlessly, as shown in Figure 12. This significantly cools the Raspberry Pi 5 CPU,, with heat analysis shown in the Engineering and Analysis section.

Figure 12: Active Cooling Fan and Heatsink Device for Raspberry Pi 5

Mounting: The protective case as shown in Figure 11 encompasses all of the components into one compact design, including the Raspberry Pi module, accelerometer, Raspberry Pi module, and connections for the camera module and LED. The case also provides management for cable connections to the power system. This case is updated from the initial prototype to include protection from dust and debris, provide cable management, and reduce electrical hazard. The case is 3D printed using PLA filament, providing a sturdy design. The case includes vents to eliminate recirculation of hot air in the case, as well as slots for the camera's attachment to the Raspberry Pi module and connections to the power system.

Camera Module: The camera module consists of a Raspberry Pi Camera Module V3 mounted on a two axis gimbal that allows for the camera angle to be adjusted based on mounting orientation. The two axis gimbal setup used two set screws to rotate the camera to the correct angle. The camera itself connects seamlessly to the Raspberry Pi 5 module via a ribbon cable, and can operate at up to 50 FPS.

Alert System: The final design utilizes an LED strip shown in Figure 13:

Figure 13: WS2812b RGB LED Strip

The LED strip connects directly to the GPIO pins on the Raspberry Pi module using cables, allowing it to become programmable. This strip is utilized by attaching it to the neck of the bandsaw, however, the length and attachment can be customized based on the customer's preference. Currently, the final design does not incorporate a baseline method to attach the LED alert system to the bandsaw, which should be included in future work for the design of the alert system.

Vibration Detection System (VDS):

The vibration detection system highlighted in the Engineering and Analysis section works using an MPU6050 sensor, which is an accelerometer that detects vibration, as shown in Figure 11. The sensor connects directly to the GPIO pins on the Raspberry Pi module using cables allowing it to become programmable. This sensor attaches directly to the connector plate using the M2.5 screws, giving a stable connection that detects vibration for the entire unit.

Power System:

The safety system uses modular battery integration to make it more suitable for the makerspace environment. The system is powered by a Milwaukee M18 XC battery, which is common in many educational makerspaces. The battery connects to a custom-designed magnetic receiver and case that incorporates a DC-DC step-down converter, shown in Figure 14, reducing the 18V supply to the 5V required by the Raspberry Pi 5 and supporting electronics. This design enables easy battery swaps without the need for rewiring or disassembly, significantly improving maintainability and uptime. The inclusion of a vibration detection system allows the device to remain dormant during idle periods, greatly reducing energy consumption and enabling the device to meet or exceed a full day of high-frequency operation on a single charge. Passive cooling through ventilation slits in the battery case ensures safe thermal performance without active cooling, maintaining reliability in demanding workshop environments.

Figure 14: SDM20-12S5, DC DC Step Down Converter

When idle, only the vibration detection system is running, the device draws approximately 1.4 amps, while during active hand detection and alert operation, it draws around 4.2 amps. Based on a typical makerspace usage cycle, the combined average current draw results in an estimated energy requirement of 10.8 amp-hours over a 6-hour operational day. Milwaukee M18 XC batteries, when stepped down to 5V, offer between 10.8Ah and 43.2Ah of usable capacity, depending on the model. This ensures that even the smallest battery in the XC line can meet a full day's use under heavy conditions, significantly improving system reliability.

How it Works:

The software developed for the bandsaw hand detection and alert generation system follows a modular, event-driven design optimized for real-time performance on embedded hardware. The system operates on a Raspberry Pi 5 and utilizes several key libraries, including *smbus2* for I2C communication with the MPU6050 motion sensor, *picamera2* for camera control, *OpenCV* for image processing and slit detection, *MediaPipe* Hands for real-time hand tracking, and *pi5neo* for controlling a NeoPixel LED strip.

Upon detecting machine vibration through the MPU6050 sensor, the Raspberry Pi initiates the camera module and begins video capture. The LED illuminates white at this point to indicate that detection is now active. At this point, an initial stabilization period of 3 seconds allows a slit detection algorithm identifying the location of the bandsaw blade and finalizing the region of interest for monitoring. This process ensures that subsequent frame analysis focuses only on relevant areas, thereby reducing computational demands.

Hand detection is continuously performed on incoming frames using MediaPipe's hand tracking model. The system evaluates the position of detected hands relative to the blade area to determine the risk level and subsequently triggers visual (LED color changes) alerts. If a hand is detected in the field of view of the camera, the LED turns green. If a hand is detected in the cutting path, the LED turns red, alerting the user of unsafe operation. Additionally, frames containing detected hands are selectively recorded to provide documentation for safety audits or incident investigations, optimizing storage efficiency by avoiding unnecessary data collection.

System Evaluation:

When measured against specifications (Table A1) defining characteristics the device needed to achieve the objective of this project, the physical components of the final design do well in key areas. The keystone of the physical alert system is the LED array, which has 144 WS2812 LEDs producing a total of over 850 lux [23], exceeding the target in specification 5. Furthermore, as detailed in the Engineering Analyses section, the device can function for at least one operational day of heavy use on the selected batteries (meeting specifications 7 and 12). While the final design sits far under the maximum weight of 51 lbs from specification 3 (with a maximum total weight of about 5 pounds), it exceeds specification 2's maximum footprint due to an oversized final iteration of the case (designed to incorporate a different LED), which, when combined with the height of the camera mount, results in a 4.2 inch total height. This is easily fixable with a case redesign, as the headroom of the Raspberry Pi inside the case can be reduced by over an inch with no adverse effects.

With the components and materials comprising the final design, the Bill of Materials (Table A2) added up to $170.44, exceeding the target of $60 set with specification 13. However, this specification was set both by using a 20% benchmark of the selling price of a low-cost band saw and incorporating a 70% margin for labor, manufacturing expenses, logistical expenses, and profit. This specification could reasonably be adjusted upwards by basing it on the median price of a bandsaw and, if possible, by lowering the incorporated margin after doing calculations to deduce the specific costs addressed in that margin.

As this final design was intended as a refined prototype and not a ready-for-market product, there are considerable savings to be made by exchanging parts for cheaper alternatives. The largest expense on the device is the Raspberry Pi at 35% of the total cost. Research into cheaper onboard computers that can be programmed to run the same system can reduce this significantly, as can research into cheaper cameras (another of the top costs). Furthermore, the DC-DC Converter, the second largest expense, was found to have alternatives at least 40% less costly. While $60 may be unrealistic without modifying the design significantly, optimizing parts for cost would make the Bill of Materials substantially leaner.

The software architecture effectively accomplishes the mission of this project, though it deviates from preliminary specifications in some ways. Though it meets sensing, response time, and processing specifications (8-10 in Table A1), the final design differed from initial conceptions in that it focused on preventing accidents by keeping users' hands out of the cutting path, without regard to distance from the blade in other directions. As a result, it fails to meet specification 6 in Table A1, as it does not alert if the minimum distance from the blade is breached from non-cutting directions. Incorporating that consideration may be a productive addition to the software should there be future improvements made, but the strategy used in the final design did comply with the intention behind the specification in that it prevents potential accidents from the user's hand being placed in the position most likely to result in catastrophic injury.

11. Manufacturing

The goal of manufacturing is to produce a product with speed, efficiency, and cost-effectiveness. A large portion of the prototyping phase for this project relies on 3D printing within the makerspace; however, this method is not practical for large-scale production due to its slow production rate. An alternative manufacturing method is injection molding, which, despite a higher initial equipment cost, enables faster production of individual units at a lower per-unit cost over time. During 3D printing, a tolerance of 0.2 mm is maintained. Injection molding is expected to achieve tighter tolerances, further improving the quality and consistency of the final product. Injection molding also supports a wide variety of materials, primarily thermoplastics, but also some thermosets and elastomers, offering flexibility in material selection for different components.

To reduce overall system costs, it is necessary to either manufacture certain components, such as the battery receiver and LED strip housing, or source these components in bulk from suppliers. The current prototype includes a Raspberry Pi 5, which constitutes the majority of the system's material cost. This component is not financially viable for large-scale manufacturing unless substantial bulk procurement discounts can be secured. As a result, future iterations of the design may consider replacing the Raspberry Pi 5 with a more cost-effective alternative, such as the ESP32-CAM, which offers adequate processing precision at a significantly lower price.

12. Risk Assessment, Safety, and Liability

Risk Assessment and Mitigation Measures:

The bandsaw hand detection and alert generation system undergoes a formal risk assessment to identify potential hazards and mitigation strategies. One major risk is false negatives, where a hand is not detected near the blade. Although unlikely due to optimized detection algorithms and validation testing, the severity is high. To address this, the system applies conservative detection thresholds and calibrates performance across varied conditions. False positives, while more likely, present minimal disruption and are mitigated through continued model refinement.

Hardware or sensor failures, including camera or motion sensor malfunction, are addressed through system monitoring and fallback alerts. User non-compliance, such as ignoring or disabling alerts, is mitigated by designing intuitive, non-intrusive signals to encourage consistent use. Processing latency is reduced by limiting the camera frame rate to 30 fps, ensuring timely alerts without overloading system resources. Privacy concerns are minimized by recording only when hands are detected and maintaining short data retention periods.

The system reduces risk by enhancing user awareness through early alerts and selective recording, allowing preventive action before incidents occur. Although injury remains possible, the design clearly positions the system as an assistive alert mechanism rather than a control device, limiting liability exposure and managing user expectations.

Impact of Risk, Safety, and Liability on Design:

Risk, safety, and liability considerations fundamentally shape the system's design. Early concepts exploring mechanical interventions, such as automatic blade braking, are rejected due to complexity, reliability concerns, and higher legal exposure. Mechanical systems also risk alienating experienced users, leading to system disablement and loss of safety benefits.

Instead, the final design employs non-intrusive alerts that enhance situational awareness without interfering with machine operation. This approach ensures broad user acceptance, providing novices with guidance while avoiding disruption for experienced operators. It also clarifies the system's role as an advisory tool rather than a guarantee of injury prevention, reducing potential legal risks.

By emphasizing assistive alerts, selective recording, and efficient event detection, the system achieves a practical balance between improving safety outcomes and encouraging continuous, reliable use across diverse user groups.

13. Patent Claims and Commercialization

There is a strong opportunity to pursue a patent for this design of a portable, computer vision-based alert system. The system's non-intrusive nature, modular battery integration, and self-calibration detection differentiate it from other existing technologies that function to protect users in workshops. Our product doesn't require invasive machine modification and has more adaptable sensing methods. Commercialization could focus initially on educational institutions and makerspaces. The current bandsaw market is valued at $3.2 billion while educational institutions purchase about 3000 units annually. In this industry, liability and safety concerns are high and there is potential to scale into broader industrial markets. Although our bill of materials exceeds our intended goal of $60, this product remains a great addition to most-commonly owned bandsaws considering that they do not contain any means of accident prevention. Investing in the safety system allows for users to exhibit safe practices while also maintaining awareness. With the bandsaw market steadily growing and increased attention to workplace safety, licensing the patent to tool manufacturers or offering the system as a retrofit kit could provide a promising business avenue.

14. Team Member Contributions

Usman - Stakeholder Matrix, Specification Sheet, House of Quality, Customer and Engineering Requirements, Industrial Design, Mockup/Prototyping, CAD model

Daniel - Specification Sheet, House of Quality, Gantt Chart, Bill of Materials, Mockup/Prototyping, Final Design, CAD model, Design and Ideation, Engineering Analyses, Executive Summary, Conclusion

Farzaan - Prior Art, Market Research, Specification Sheet, Customer and Engineering Requirements, Mockup/Prototyping, CAD model, Patent and Commercialization, Introduction

Krish - Executive Summary, Design and Ideation, Introduction, Software Development

Tyler - Stakeholder Matrix, Manufacturing

Huansang - Market Research, Software Development, Final Design, Risk Assessment

15. Summary and Future Work

The presented information serves to illustrate the team's process of narrowing the scope of the project to a defined design concept and pursuing this idea through the prototyping process. The timeline of the project (outlined in the Gantt Chart listed as Figure A3 in the Appendix), largely progressed as intended, though prototyping began more gradually then can be represented by the chart.

From the analysis phase, the team converged on a specific machine focus (bandsaw), a specific risk focus (laceration and sever injuries), and modes of fulfilling the functions deemed necessary to solve the design problem as defined by the two focuses. Execution of the design as defined during the analysis phase included designing a mounting module for the device, obtaining parts and fabricating an initial prototype, improving upon the initial prototype to the final design, and creating functioning code for hazard recognition and warning. Software challenges that were addressed include self-calibration, a tiered alert system, and active power management, while the device's precision and processing speed were refined significantly to meet target specifications. Improvements in the hardware aspect included redesign of the mounting module to reduce size and incorporate cooling and dust shielding, though optimization of cost remains as a necessary step in future work.

Administrative challenges with work-distribution structure and intra-team communication were present near the inception of the project but improved over the course of the team's work. The rate at which the device improves in its ability to meet target specifications in the weeks approaching the Design Expo was closely monitored, and though the team was prepared to make logistical adjustments in team administration to ensure we were on pace to present a high-level product at the conclusion of this project, few were needed. We thank our sponsor and associated advisors for their timely and helpful communication, as their input and guidance was invaluable in both the execution of the project and the administration of the team.

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Appendix

Table A1: Full Specification Sheet

			Specification Sheet For Makerspace Safety System	Issued: 4/25/2025			
No.	Date	D/W	Requirements	Source	How Validated	Responsible	Link
			Physical Characteristics				
1	3/8/2025	W	Device edges have 2mm radius fillet	ANSI/HFES 100-2007 Section 6.2.1.7	CAD model/Physical measurement	Usman	Reference 2
2	3/8/2025	D	Device footprint is at most 12"x4"x4" (LxWxH)	Minimum Dimensions for 1.5+ HP Bandsaws on Market	CAD model/Physical measurement	Usman/Daniel	Reference 3
3	3/8/2025	D	Device/System is under 51 pounds	NIOSH Lifting Equation with 2 Hands	Physical measurement	Usman/Daniel	Reference 5
			Operation				
4	3/8/2025	W	Auditory sensing alerts are at least 110 dB	ISO 7731 Ergonomics	Auditory component specifications	Tyler	Reference 14
5	3/8/2025	D	Device is capable of producing at least 750 lux (or 85 lumens) of light for safety	OSHA Factories and Workshops Lighting Requirement	LED component specifications	Tyler	Reference 15
6	3/10/2025	D	Safety bounding box maintains a minimum of 3" clearance around blade in all directio	Kansas City Woodworkers Guild Safety Guide	Physical testing	Krish/Helen	Reference 16
7	3/8/2025	D	Device has continous power for at least 6 hours of high-frequency use	Invention Studio hours	Battery specification/USB power meter testing	Farzaan/Helen/Daniel	Reference 22
			Sensing				
8	3/11/2025	D	Device is capable of continously sensing unsafe operation up to 12 inches away	Doall bandsaw dimensions (from a GT facility)	Physical testing	Krish/Helen	Reference 17
9	3/8/2025	D	Device has a total response time of less than 109 ms	Minimum human reaction time data	Physical testing/Observation	Krish/Helen	Rererence 18
10	3/8/2025	D	Device processes a maximum of 50 FPS	Raspberry Pi Camera 3 Specification	Camera specifications	Krish/Helen	Reference 19
			Human Factors				
11	3/8/2025	D	Device has no exposed electrical componenents.	IEC 60204-1 (Safety of machinery - Electrical equipment of machine	Physical inspection	Usman/Daniel/Farzaan	Reference 20
12	3/8/2025	W	Device needs to be maintained/recharged at most once every 6 hours	Invention Studio hours, Makerspace guidelines	Battery specification/USB power meter testing	Helen/Farzaan	Reference 22
			Miscellanous				
13	3/8/2025	W	BOM cost can be at most $60	Market Research (30% of 20% of Cheapest 1.5 HP Band Saw)	BOM	All members	Reference 21

Extra Links	
https://aeasseincludes.assp.org/professionalsafety/pastissues/064/12/F3Weaver_12	Describes safety risks and associated results
https://woodgears.ca/bandsaw/plans/specifications.html	General bandsaw dimensions for reference
https://cdn.sparkfun.com/assets/9/9/0/d/e/raspberry-pi-5-product-brief.pdf	Raspberry Pi 5 Specification
https://sawmillcreek.org/showthread.php?200942-Noise-level-readings-in-the-shop	Ambient noise level of a bandsaw

Notes:
Used 13"x13" cutting surface of smallest 1.5+ HP band saw on market to convert lux to lumens

Table A2: Bill of Materials

Part	Model Name	Source	Quantity	Unit Cost
Onboard Computer	Raspberry Pi 5, 4GB RAM	Raspberry Pi	1	$60.00
Camera	Camera Module 3	Raspberry Pi	1	$24.99
Active Cooler	Active Cooler	Raspberry Pi	1	$9.50
M2.5 Screw	M2.5 Screw	Helifouner, Amazon	16	$0.01
M2.5 Nut	M2.5 Nut	Helifouner, Amazon	12	$0.01
M2.5 Short Standoff	M2.5 Short Standoff	Helifouner, Amazon	4	$0.02
M2.5 Long Standoff	M2.5 Long Standoff	Helifouner, Amazon	4	$0.02
Accelerometer	MPU6050	Huarew, Amazon	1	$3.33
Power Cord	USB-C to 2-pin Bare Wire Open End Power Cord	Maixbomr, Amazon	1	$4.50
LED	WS2812 B RGB LED Strip, 3.2 ft DC 5V	KWMSTPLT, Amazon	1	$12.99
DC-DC Converter	SDM20-12S5	Mean Well USA, DigiKey	1	$30.07
Battery Receiver	Power Wheel Adapter for Milwaukee M18 Battery	ChoiKWong, Amazon	1	$8.00
Magnets	30 lb Bolt-on Magnets, M6 Bolts	MAGXCENE, Amazon	4	$3.49
Set Screw	#6x3/16 Set Screw	Cap, Home Depot	2	$1.13
PLA	ELEGOO PLA Filament 1.75 mm	ELEGOO, Amazon	200 g	$13.99 (per kg)
			Total Cost	**$170.44**

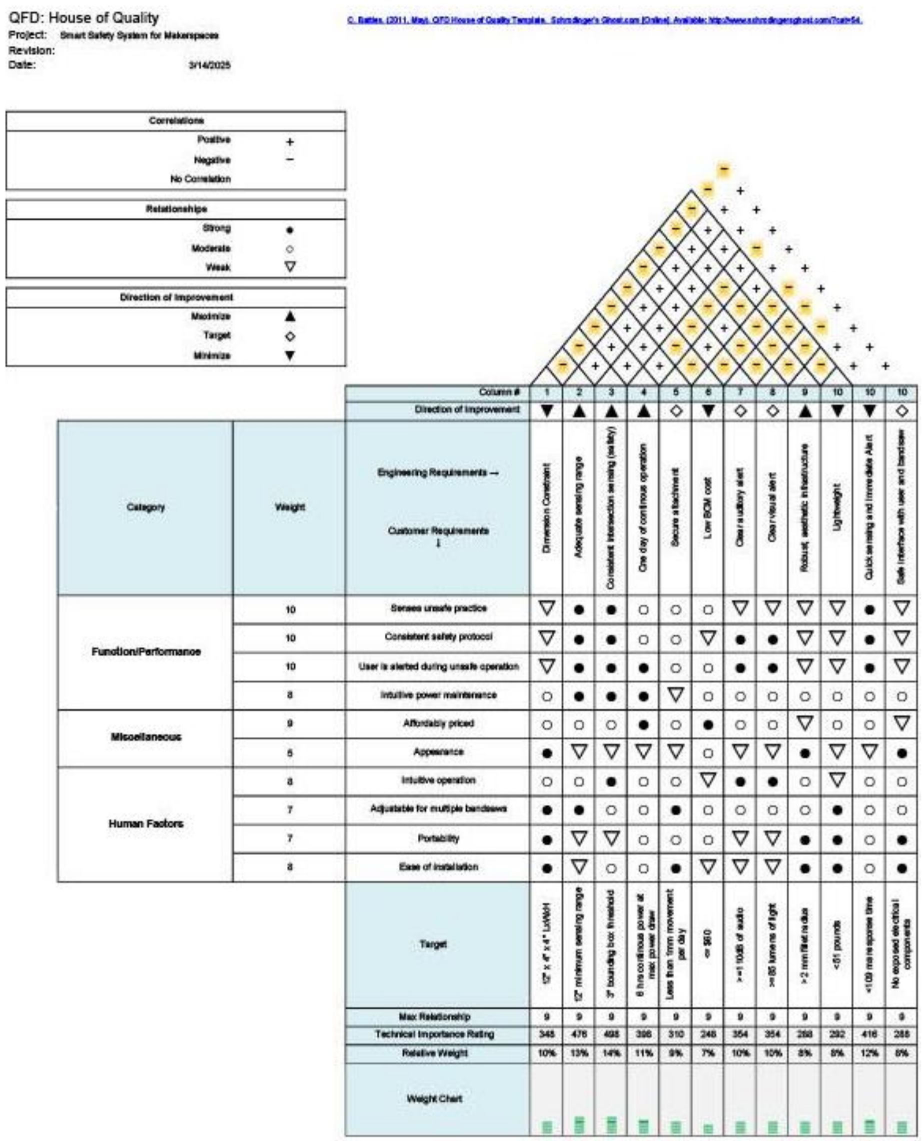

Figure A1: Full House of Quality

Categories/Function	Option 1	Option 2	Option 3
Check Alignment	 Recognize Reference Point on Cutting Surface	 Hardware-Based Position/Orientation Sensing	
Minimize External Signal Noise	Software (Signal Processing Techniques)	Hardware (Insulate sensors from interference)	
Mount on Machine Body	Removable Hardware 	Magnetism 	Adhesive/Velcro 
Connect to Power Source	 External Source	 Incorporated Source	
Alert User of System Condition	 Visual Cue of Arming Status	 Visual Cue	

Figure A2: A Morphological Chart with design solutions for device functions (continued below)

		of Power Status	Audible Cue of Power Status
Recognize Hand	Infrared Sensor 	Ultrasonic Sensor 	Computer Vision 
Recognize Intersection with Cutting Path	Infrared Sensor 	Ultrasonic Sensor 	Computer Vision 
Create Visual Cue	Light 	Display 	
Create Auditory Cue	Electronic 	Mechanical 	

Figure A2 (continued)

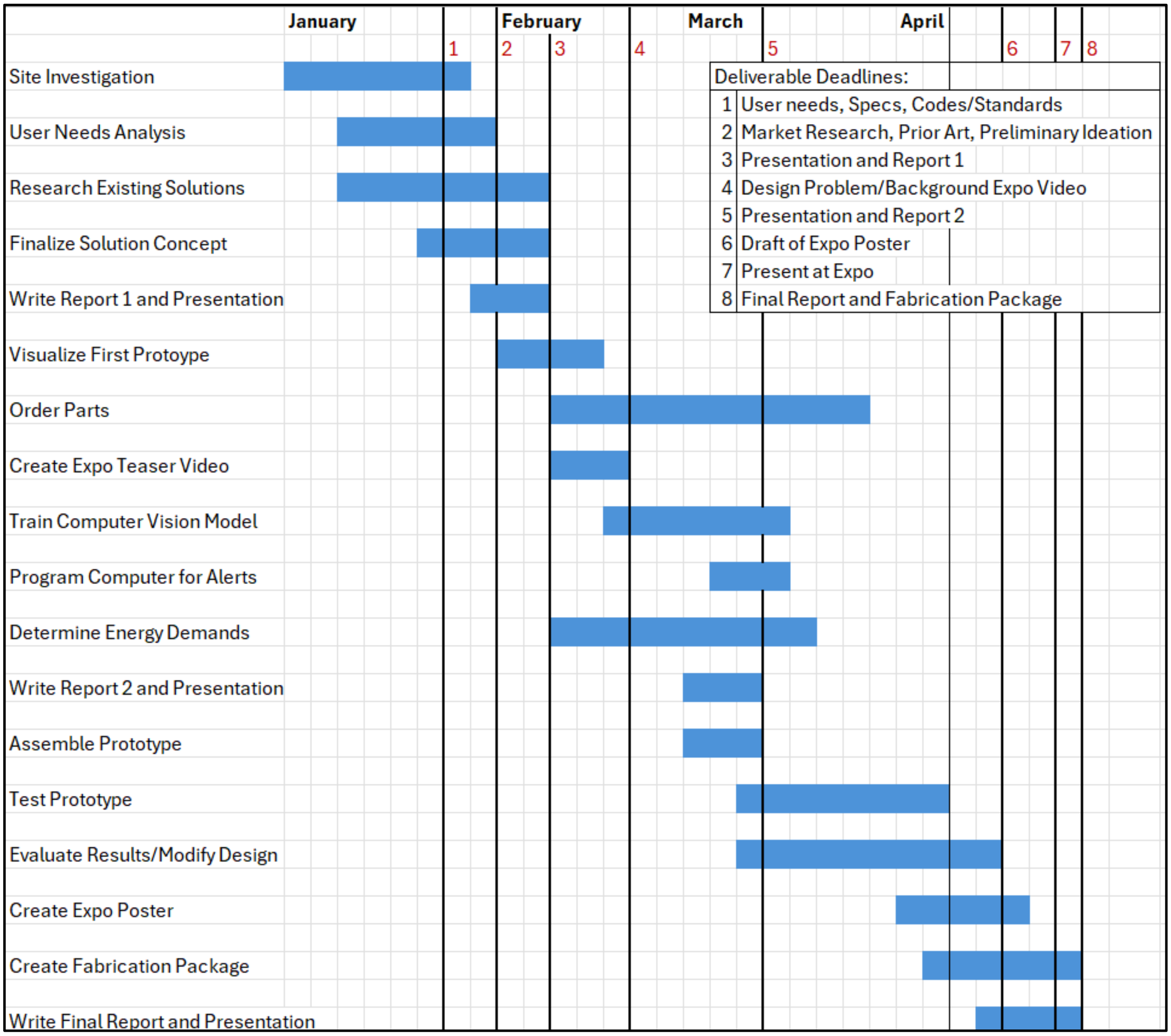

Figure A3: Gantt chart showing the project plan as it is now predicted to progress